

Case Study 2: Counterfactual Fairness in Employee Attrition

A Causal Fairness Analysis using DoWhy

Dataset: IBM HR Analytics Employee Attrition (Kaggle) | June 2026

1. Objective

This case study investigates whether mandatory overtime causally drives employee attrition, and whether this effect is equitable across genders. Using counterfactual fairness analysis, we ask: would an employee's attrition outcome change if their gender were different, everything else held constant? If yes, the overtime policy is causally unfair.

This case study also demonstrates the Explainability Plane from Bareinboim's framework -- XAI methods like SHAP operate at the associational layer (correlations), while counterfactual fairness operates at the counterfactual layer (the highest rung of Pearl's Causal Hierarchy). This is the conceptual bridge between standard explainability and causal reasoning.

Variables

Treatment (T): Overtime assignment (OverTime = 1)

Outcome (Y): Employee attrition (Attrition = 1)

Protected Attribute: Gender (Male = 1, Female = 0)

Confounders: Age, JobLevel, MonthlyIncome, JobSatisfaction | Dataset size: 1,470 records

2. Exploratory Analysis

Naive comparison of attrition and overtime rates across gender groups. These observations are purely associational and do not establish causality.

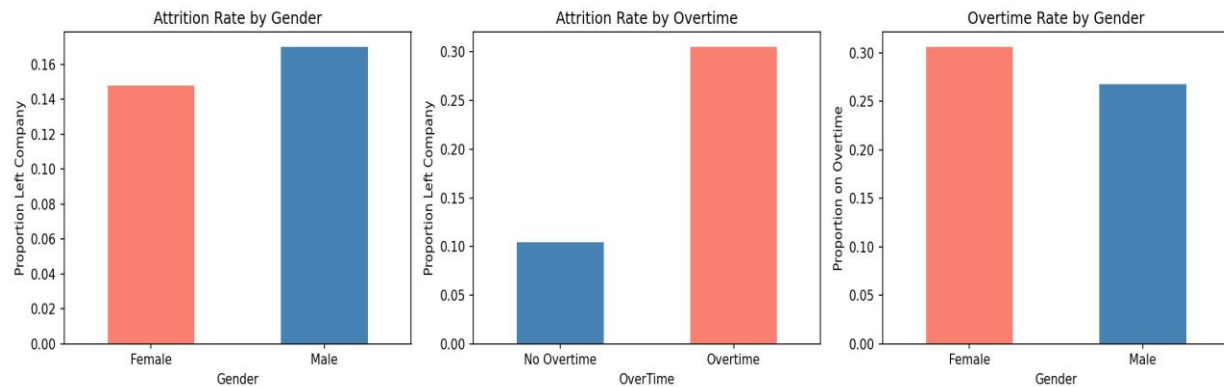


Figure 1: Attrition by gender (left), attrition by overtime (middle), overtime rate by gender (right)

Key observations: Males have slightly higher attrition (17%) than females (15%). Overtime workers have 3x higher attrition (30% vs 10%), a strong signal. Notably, females are assigned overtime more frequently than males (30% vs 27%), which motivates the fairness investigation -- are women being pushed out of the company at higher rates due to disproportionate overtime assignment?

3. Causal Graph (DAG)

Gender is included as a confounder in the DAG -- it influences both who gets assigned overtime and who ultimately leaves the company. This structure allows us to later isolate gender-specific causal effects and test for differential impact.

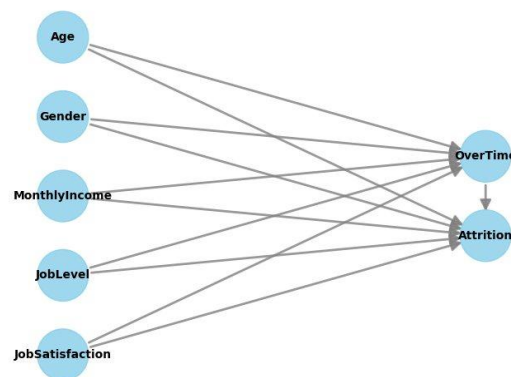


Figure 2: Causal DAG -- Gender and other confounders create backdoor paths between OverTime and Attrition

4. Causal Identification

DoWhy confirmed the backdoor estimand -- conditioning on {Age, Gender, JobLevel, MonthlyIncome, JobSatisfaction} satisfies the backdoor criterion and makes the causal effect of overtime on attrition identifiable:

```
E[Attrition | do(OverTime)] = d/d[OverTime] E[Attrition | Age, Gender,
JobLevel, MonthlyIncome, JobSatisfaction]
```

5. Causal Effect Estimation

The Average Treatment Effect (ATE) of overtime on attrition was estimated using linear regression adjustment on the backdoor variables.

```
ATE = 0.207 (p-value: 6.5e-24)
Naive correlation effect = 0.200
```

Unlike Case Study 1, where confounders explained most of the naive effect, here the causal estimate holds up strongly after adjustment. Overtime causally increases attrition by ~21 percentage points with extremely high statistical significance. The effect is real and robust.

6. Counterfactual Fairness Analysis

We estimate the causal effect of overtime on attrition separately for male and female subgroups. A counterfactually fair policy requires that this effect be equal across groups -- i.e., overtime should not be more damaging to one gender than the other.

```
Female ATE: 0.2069
Male ATE: 0.2069
Fairness gap: 0.0000
```

The causal effect of overtime on attrition is identical for both genders. Overtime policy passes the counterfactual fairness test -- it does not differentially harm one gender over the other.

7. Causal Effect of Gender on Overtime Assignment

A secondary fairness question: does gender causally determine who gets assigned overtime, after controlling for job-related factors? We build a second causal model with gender as treatment and overtime as outcome.

```
Causal effect of Male gender on Overtime: -0.039 (p =
0.103)
```

The negative coefficient means being male slightly reduces the probability of overtime assignment, but the effect is not statistically significant ($p = 0.10$). The gender gap in overtime rates observed in the EDA is explained by confounders like job level and income -- not by direct gender-based discrimination in assignment.

8. Refutation Tests

Random Common Cause Test: Estimate moved from 0.2069 to 0.2068 -- negligible change, confirming stability.

Placebo Treatment Test: Estimate collapsed to -0.0015 when treatment was randomized, confirming the original estimate captures a genuine causal signal.

9. Summary of Results

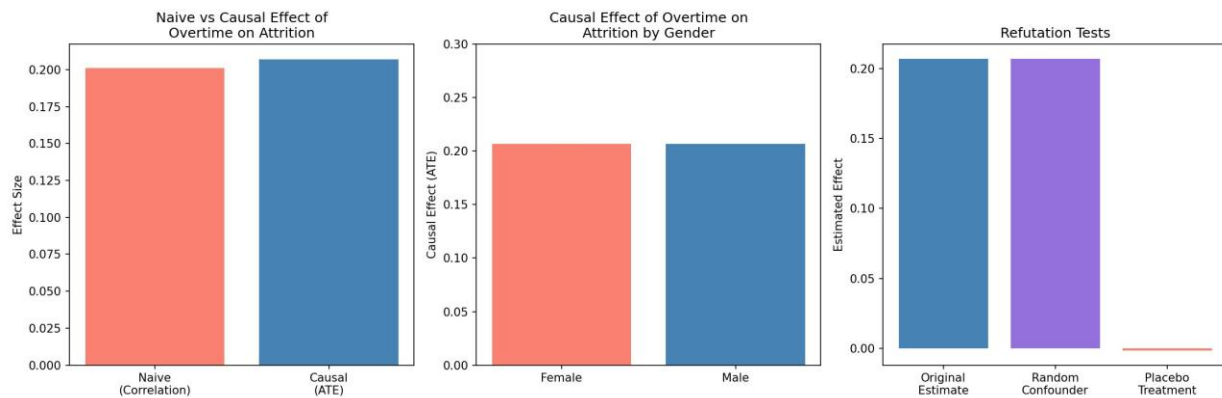


Figure 3: Naive vs causal effect (left), gender-stratified causal effect (middle), refutation results (right)

10. Conclusion

This case study demonstrates two complementary findings. First, overtime is a genuine causal driver of employee attrition -- a 21 percentage point increase in attrition probability, robust to all refutation tests. This is a practically significant result for HR policy: reducing mandatory overtime would meaningfully reduce turnover.

Second, the overtime policy is counterfactually fair -- the causal effect is identical for male and female employees (0.2069 each), and there is no statistically significant gender-based discrimination in overtime assignment after controlling for job-related variables. The apparent gender gap in overtime rates in the raw data was confounded.

Key concepts demonstrated:

- Counterfactual fairness and gender-stratified causal effect estimation
- Path-specific effects and protected attribute analysis
- Explainability Plane: associational (XAI/SHAP) vs counterfactual (causal fairness) reasoning
- Backdoor criterion and ATE estimation in an HR analytics context
- Refutation testing for robustness validation

The contrast between Case Study 1 and Case Study 2 is itself instructive: in CS1, causal analysis revealed that a strong correlation was mostly confounded; in CS2, causal analysis confirmed that a strong correlation reflects a genuine causal effect. This demonstrates that causal methods are not biased toward any particular conclusion -- they reveal the truth in the data regardless of direction.